Problem Sheet — IMC Training

May 27, 2026

Problem 1. (IMC 2008, Day 2, Problem 2) Two different ellipses are given. One focus of the first ellipse coincides with one focus of the second ellipse. Prove that the ellipses have at most two points in common.

Problem 2. (IMC 2009, Day 1, Problem 1b) Suppose that f and g are continuous real-valued functions on the real line and $f(r) \le g(r)$ for every rational r. Prove that $f(x) \le g(x)$ for every real x.

Problem 3. (IMC 2009, Day 1, Problem 3) In a town every two residents who are not friends have a friend in common, and no one is a friend of everyone else. Let us number the residents from 1 to n and let a_i be the number of friends of the i-th resident. Suppose that $\sum_{i=1}^n a_i^2 = n^2 - n$. Let k be the smallest number of residents (at least three) who can be seated at a round table in such a way that any two neighbors are friends. Determine all possible values of k.

Problem 4. (IMC 2009, Day 2, Problem 2) Suppose $f : \mathbb{R} \to \mathbb{R}$ is a twice differentiable function satisfying $f(0) = 1$, $f'(0) = 0$, and for all $x \in [0, \infty)$,

$$f''(x) - 5f'(x) + 6f(x) \ge 0.$$

Prove that for all $x \in [0, \infty)$,

$$f(x) \ge 3e^{2x} - 2e^{3x}.$$

Problem 5. (IMC 2008, Day 2, Problem 1) Let n, k be positive integers and suppose that the polynomial $x^{2k} - x^k + 1$ divides $x^{2n} + x^n + 1$. Prove that $x^{2k} + x^k + 1$ divides $x^{2n} + x^n + 1$.

Problem 6. (IMC 2009, Day 2, Problem 3) Let $A, B \in M_n(\mathbb{C})$ be two $n \times n$ matrices such that

$$A^2B + BA^2 = 2ABA.$$

Prove that there exists a positive integer k such that $(AB - BA)^k = 0$.

Problem 7. (IMC 2008, Day 1, Problem 6) For a permutation $\sigma = (i_1, i_2, \dots, i_n)$ of $(1, 2, \dots, n)$ define

$$D(\sigma) = \sum_{k=1}^{n} |i_k - k|.$$

Let $Q(n, d)$ be the number of permutations σ of $(1, 2, \dots, n)$ with $d = D(\sigma)$. Prove that $Q(n, d)$ is even for $d \ge 2n$.

Problem 8. (IMC 2021, Day 2, Problem 5) Let A be a real $n \times n$ matrix and suppose that for every positive integer m there exists a real symmetric matrix B such that

$$2021\, B = A^m + B^2.$$

Prove that $|\det A| \le 1$.

Problem 9. (IMC 2009, Day 1, Problem 4) Let $p(z) = a_0 + a_1 z + a_2 z^2 + \cdots + a_n z^n$ be a complex polynomial. Suppose that $1 = c_0 \ge c_1 \ge \cdots \ge c_n \ge 0$ is a sequence of real numbers which is convex (i.e. $2c_k \le c_{k-1} + c_{k+1}$ for every $k = 1, 2, \dots, n-1$), and consider the polynomial

$$q(z) = c_0 a_0 + c_1 a_1 z + c_2 a_2 z^2 + \cdots + c_n a_n z^n.$$

Prove that

$$\max_{|z| \le 1} |q(z)| \le \max_{|z| \le 1} |p(z)|.$$

Problem 10. (IMC 2021, Day 2, Problem 6) For a prime number p, let $\mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z})$ be the group of invertible 2×2 matrices of residues modulo p, and let S_p be the symmetric group on p elements. Show that there is no injective group homomorphism $\varphi : \mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z}) \to S_p$.